



Deleterious Effects of Polypropylene Microplastic Ingestion in Nile Tilapia (*Oreochromis niloticus*)

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Abstract

The effect of daily ingestion of polypropylene microplastic on the health of tilapia, *Oreochromis niloticus*, was evaluated. 60 fish (± 200 g) were placed in 6 aquariums ($n=10$, 100 L each), constituting the following treatments: Control (without the addition of polymer), fed with 100 and 500 μg of polypropylene/kg of body weight (b.w.), respectively. After 30 days of feeding, the animals were submitted to blood collection for hemogram and biochemical study and later euthanized for gut microbiological analysis, somatic index of liver, spleen, heart, kidney, stomach, and intestine. In the serum biochemical study, an increase in cholesterol and serum Aspartate Aminotransferase (AST) activity levels was observed in animals treated with 500 μg of polypropylene. Tilapia-fed polypropylene in the diet showed an increase in thrombocyte and total leukocyte counts, marked by a significant increase in the number of circulating lymphocytes. The results of the somatic study revealed a significant increase in the stomach, liver, and heart of tilapia fed with the polymer. Increase in the number of Gram-negative microorganisms and decrease in mesophilic aerobic microorganisms were observed in the gut of fish exposed to the polymer, including a dose-response effect was observed for these analyses. Therefore, tilapias fed daily with diets containing polypropylene for 30 consecutive days showed deleterious effects, resulting in systemic inflammatory disturbs by altering liver functions, leukocyte profile, and organ morphometry, as well as changes in the intestinal microbiota. Such results demonstrate the impairment of fish health, highlighting the need for further studies that evaluate the impact of microplastics on aquatic organisms.

Keywords Waste · Aquatic ecosystems · Aquaculture · Environmental pollution · Fish

Introduction

Polypropylene is used in the production of films for packaging, appliance cabinets, and luggage, when not recycled polypropylene suffers degradation in the environment, forming microparticles by processes such as mechanical abrasion, UV radiation, and microbiological degradation (Ivleva et al., 2016). Successive fragmentation generates increasingly smaller debris, the identification of microplastic

fragments first occurred in the 70s, but only in 2004, the term microplastics (MPs) was used to refer to plastics with a size between 0.001 and 5 mm (Olivatto et al. 2018). Studies point to the presence of polypropylene microplastic residues found in coastal and marine environments, demonstrating concern about the ingestion of these polymers at various levels of the food chain in aquatic environments (Shim and Thomposon, 2015). Ingestion of microplastics by fish widely occurs in various natural aquatic environments, and exposure to them could cause health problems. In a review realized by Wang et al. (2020), the toxic effects on fish are diverse, being associated with a complex condition that involves the fish species susceptibility, the plastic polymer composition, the particle size, as well as the time and form of exposure (through the water or food). Genotoxicity and histological changes were described in tilapias, *Oreochromis mossambicus*, fed with 100, 500, and 1000 mg of

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polypropylene microplastics /kg for acute exposure by 96 h (Jeyavani et al. 2023). According to these authors, fish fed with the polymer for a longer period of time increased its toxicity (sub-acute exposure for 14 days).

Veterinary clinical pathology studies involving serum biochemical analyzes of liver and kidney function, as well as hematological studies help in the diagnosis of morbid conditions (Aracati et al. 2021). Morphometric changes in the somatic index of organs can represent adaptive strategies for the conservation of some physiological functions or endpoints to evaluate acute or chronic exposure to a chemical present in water and sediment (Liebel et al. 2013). According to Yukgehnaish et al. (2020), the composition and interactions of the gut microbiota play an important role in the metabolism of dietary substrates and immune system modulation, directly related to animal health.

Fish represent one of the most important bioindicators of water quality, and some fish species are sensitive to various changes in the environment, such as pH, temperature, electrical conductivity, chemical compounds, among others (Moraes et al. 2022). In this context, tilapia stands out as the third species of freshwater fish created in the world, representing 9% with about 4,407.2 thousand tonnes (FAO, 2022), since they are fast-growing animals, good fillet yield and high acceptability by consumers. Based on the importance of microplastics in the aquatic environment, the aim of this study was to evaluate the effects of ingestion of 100 and 500 µg of polypropylene/kg of b.w. for thirty days on the health of Nile Tilapia, *Oreochromis niloticus*.

Materials and Methods

Sixty Nile tilapias (200 ± 5 g) from the same spawning were randomly distributed in 6 aquariums ($n = 10$), with a capacity of 100 L of water each, supplied with running water, from an artesian well, with a flow of 1 L /min, to constitute three treatments (with two aquariums each): control, 100 and 500 µg of microplastic/kg of b.w. After transport from Aquabel Fish Farm (Porto Ferreira, São Paulo State, Brazil) to the aquariums, in the first three days were administered NaCl at a concentration of 6.0 g/L to reduce the stress (Carneiro and Urbinati 2001). The fish were acclimated for 15 days so that plasma cortisol levels and osmolarity returned to baseline levels. Feeding was carried out twice a day, corresponding to 2% of the aquarium biomass. The water quality was determined at the time of feeding, with the temperature and dissolved oxygen concentration measured by the YSI device, model 55, and the pH and electrical conductivity by the YSI device, model 63. Their values remained within the appropriate range for the well-being of tropical fish (dissolved oxygen = 4.07 ± 0.89 mg L⁻¹;

temperature = 27.64 ± 2.05 °C; pH = 7.64 ± 0.54 ; and conductivity = 208.29 ± 97.57 µS/cm). This experimental trial was carried out at the Laboratory of Animal Pharmacology and Toxicology, Brazil University, Descalvado/SP, and conducted in accordance with the ethical principles in animal experimentation, approved by the ethics committee for the use of animals (CEUA) under No. 2,200,039.

The extruded commercial feed (NUTRIPISCIS® - PRESENCE Company, containing 36% crude protein, 12% moisture, 70 g/kg ether extract, 140 g/kg mineral matter, 50 g/kg crude fiber, 50 g/kg calcium 25 g/kg, phosphorus 8 g/kg and vitamin C 350 mg/kg was used to compose the experimental diets. For the preparation of the diets, the feed was weighed daily in proportion to the tilapia's average weight in each aquarium. Then, the doses of 100 and 500 µg of polypropylene /kg⁻¹ bw were determined, which were previously homogenized in 2% of vegetable oil, to be later mixed in the feed, ensuring homogeneous distribution throughout the diet. To standardize the diets and nutritional balance, 2% vegetable oil was added to the diet of the control group. Feeding was carried out twice a day (8 am and 5 pm), with the administration of 2% of the aquarium biomass for a period of 30 days. For analysis of polypropylene microplastic (Braskem, Brazil), samples were placed on glass slides and analyzed by optical microscopy (Carl Zeiss Jena, Germany). Fourteen (fourteen) fields were randomly selected and the images captured by Opton CMOS TA-0124-D to determine the area and perimeter of microplastic particles using Image-J software (Schneider et al. 2012). Mean values and range of variation revealed that microplastic particles in the diet had an area of $0.91 \mu\text{m}^2$ (0.05 to $5.43 \mu\text{m}^2$) and perimeter $3.43 \mu\text{m}$ (0.89 to $9.82 \mu\text{m}$).

After the 30-day exposure period, the animals were euthanized for clinical trials. Fish were anesthetized by immersion in an aqueous solution of benzocaine at a rate of 10,000 mg/liter for capture and 500 mg/liter for euthanasia. 3mL of blood was collected from the caudal vessel with the aid of disposable syringes by puncturing the caudal vessel. Blood were sampling and aliquoted into two sets: one syringe coated with heparin (5000 IU) and one without heparin, to obtain plasma (1mL) and serum (2mL), respectively. The determinations of global counts of red and white cells using plasma were carried out in a Neubauer chamber, using Natt and Herrick's solution, with diluent in the proportion of 1:100. The percentage of hematocrit was determined in a microcentrifuge and the amount of circulating hemoglobin with Drabkin's reagent for reading at a wavelength of 540 nm. Mean corpuscular volume (MCV) and mean corpuscular hemoglobin concentration (MCHC) were obtained by calculating $\text{MCV} = (\text{HT} / \text{HE}) * 100$ and $\text{MCHC} = (\text{HG} / \text{HT}) * 100$, respectively. The differential leukocyte count was performed in blood smears with a count

of 200 cells, establishing the percentage of each cell type of interest, after staining with May-Grünwald-Giemsa-Wright (Farias et al. 2016).

Blood samples collected without anticoagulant were centrifuged at 5,000 rpm for five minutes to obtain serum. The biochemical study of serum aspartate aminotransferase (AST), alanine aminotransferase (ALT), alkaline phosphatase (AP), total proteins, albumin, glucose, cholesterol, triglycerides, and creatinine was determined by colorimetric and enzymatic processes in a semiautomatic biochemical analyzer (Model LABQUEST – BIOPLUS) following protocol described by Costa et al. (2022).

The liver, spleen, heart, kidney, stomach, and intestine were collected to evaluate the somatic index, which was performed according to Weibel et al. (1969), by the ratio between organ weight (OW) and body weight (BW) expressed as somatic index by the formula: Somatic index (%IS) = $OW \times 100 / BW$.

In order to carry out the microbiological analyzes, the intestines of five fish from each treatment were collected. This material was kept in refrigeration and immediately sent to the Laboratory of Microbiology for analysis. For the cultivation of intestinal microbiota, agar media were used: Trypticase Soy Agar (TSA), Blood Agar, McConckey, Salmonella-Shigella (SSA), Eosin-Methylene Blue (EMB), Sabouraud-Dextrose and Selenite-Cysteine Broth. The material for cultivation was obtained with sterile “swabs”, rubbed five times over the material, and deposited in tubes containing 10 mL of sterile peptone water (0.1%), homogenized for three minutes (vortex). The solution obtained was used for cultivation in selenite-cysteine broth (1.0mL was used) and for serial dilutions (10 – 1 – 10 – 3). Serial dilutions (0.1mL) were used for inoculation on agar media: Trypticase Soy Agar (TSA), Blood Agar, McConckey, Eosin-Methylene Blue (EMB), Sabouraud-Dextrose. The Petri dishes were incubated in aerobic conditions at 37°C for 24/48 h. The cultivation of anaerobes was carried out on a blood agar medium. At the same temperature and time. An anaerobic jar was used, and Gaspak Ez Anaerobe (BBL®) as an anaerobic generator. All experiments were performed in triplicate. The selenite-cysteine broth was incubated at 37°C for 18 h, under agitation (225 rpm), when 0.1mL was inoculated in Petri plates containing Salmonella/Shigella AGAR, incubated at 37°C for 24/48 h. After the incubation period, the growth and morphological characteristics of the colonies were verified and their counts were performed. Colonies were picked for identification by Gram staining and biochemical tests. Gram-positive bacteria were submitted to coagulase, hemolysis, and DNAase tests, while Gram-negative bacteria were evaluated using the API20E system (BIOMÉRIEUX®).

The experimental design was completely randomized with three treatments. Blood count, leukogram, biochemistry, somatic analysis of organs, and intestinal microbiota data were analyzed using the Minitab program (2018), using analysis of variance (ANOVA), normality of errors (Cramer-Von-Mises) and homoscedasticity of variance (Light). When statistical significance was observed, Fisher's LSD test ($P < 0.01$; 0.05; 0.10) was applied to compare means and standard error of means (SE).

Results and Discussion

The results observed in the erythrocyte counts, percentage values of hematocrit, hemoglobin, MCV, and MCHC did not reveal significant statistical differences ($p \geq 0.05$) between control fish and those treated with the two doses of microplastic 100 and 500 µg of polypropylene (Table 1). However, there was a significant increase ($P < 0.01$) in the total leukocyte, thrombocyte, and lymphocyte counts of fish fed with the polymer in the diet when compared to control fish. The fish monocyte and neutrophil counts did not show significant statistical differences ($p \geq 0.05$) between the individuals who received the different treatments (Table 1). The increase in the number of lymphocytes, total leukocytes, and thrombocytes in animals fed with the polymer suggest the hypothesis of inflammatory alterations by the polypropylene harmful action in the gastrointestinal tract of tilapia. Bhagat et al. (2020) found that lesions in the intestine of zebrafish (*Danio rerio*) due to the presence of MPs caused direct effects on the immune system of fish. For Hirt and Body-Malapel (2020), MPs interact with the fish intestine and can move into the bloodstream causing systemic disorders.

According to Kim et al. (2020), the response against substances such as microplastics can be interpreted by the immune system in several ways. Contact with 100 and 1000 µg of polyethylene microplastic/L resulted in a significant increase in lysozyme levels associated with an increase in the number of circulating neutrophils in *Danio rerio* (Limonta et al. 2019).

The serum biochemical study of tilapia revealed a significant increase in serum enzymatic activity of AST and ALT in fish exposed to the polymer, with this effect being dose-response dependent (Table 2). According to Kim and Kang (2016), AST, ALT and ALP are excellent indicators of liver disorders in fish. Increased serum activity of these enzymes reveals pathological lesions in cell membrane permeability and/or cytotoxicity of hepatocytes, suggesting the deleterious action of polypropylene microplastics on liver integrity, corroborating the findings by Hamed et al. (2019) who observed a significant increase in juvenile *O. niloticus* when

Table 1 Mean¹ (± SE) and ANOVA² values observed in the hematological analysis of tilapia treated with polypropylene

Treatment ³	Erythrocytes (10 ⁶ /mm ³)	Hemoglobin (g/dL)	Hematocrit (%)	MCV ⁵ (fL)	MCHC ⁵ (g/dL)
Control	1,51 ± 0,04	5,43 ± 0,18	22,72 ± 0,73	148,31 ± 4,92	24,49 ± 1,08
100 µg	1,43 ± 0,04	5,65 ± 0,23	21,75 ± 0,51	143,84 ± 4,09	26,19 ± 1,20
500 µg	1,54 ± 0,07	5,77 ± 0,31	21,61 ± 0,73	148,31 ± 7,68	28,51 ± 1,99
P Value	0,49	0,76	0,63	0,92	0,33
F Value	0,71	0,28	0,47	0,09	1,12
Treatment ³	Leukocytes (µL)	Thrombocytes (µL)	Monocytes (µL)	Lymphocytes (µL)	Neutrophils (µL)
Control	41,747 ± 2699 ^B	40,502 ± 2448 ^B	1995 ± 277	34,665 ± 2438 ^B	5087 ± 608
100 µg	47,512 ± 2784 ^B	41,237 ± 2088 ^B	2203 ± 193	38,282 ± 1965 ^B	5157 ± 512
500 µg	59,291 ± 3117 ^A	52,858 ± 2842 ^A	1932 ± 288	51,091 ± 2825 ^A	6060 ± 807
P Value	0,003***	0,010***	0,777	0,001***	0,627
F Value	6,47	5,00	0,25	8,29	0,47

¹Means (n = 10) followed by the same letter do not differ by the Fisher test (P < 0.01–0.05–0.10)²Analysis of variance represented by capital letters compare the different treatments in the column³Control - animals not treated with Polypropylene; Treatment with 100 or 500 µg of Polypropylene

*significant for P < 0.10; **significant for P < 0.05; ***Significant for P < 0.01

Table 2 Mean values¹ (± SE) and ANOVA² observed in the serum biochemical analyzes of AST, ALT, ALP, albumin, cholesterol, triglycerides, total proteins, creatinine, and glycemia of tilapia fed with polypropylene

Treatment ³	ALT (U/L)	ALP (U/L)	AST (U/L)	Cholesterol (g/dL)	
Control	8,50 ± 0,88 ^B	22,68 ± 2,39	89,80 ± 3,67 ^C	114,33 ± 8,27 ^B	
100 µg	11,13 ± 0,30 ^B	25,79 ± 6,44	154,38 ± 4,12 ^B	117,41 ± 9,80 ^B	
500 µg	19,90 ± 0,81 ^A	29,40 ± 1,00	226,93 ± 12,23 ^A	147,93 ± 7,93 ^A	
P Value	0,001***	0,300	0,001***	0,019**	
F Value	3,14	1,56	18,05	4,56	
Treatment ³	Triglycerides (pg)	Total protein (g/dL)	Creatinine (mg/dL)	Glycemia (mg/dL)	Albumin (g/dL)
Control	77,20 ± 11,73	2,79 ± 0,18	0,89 ± 0,03 ^A	56,23 ± 1,49	1,19 ± 0,07
100 µg	72,63 ± 7,84	2,74 ± 0,13	0,76 ± 0,04 ^{AB}	55,50 ± 2,82	1,38 ± 0,09
500 µg	108,26 ± 9,90	3,07 ± 0,13	0,63 ± 0,04 ^B	62,05 ± 3,75	1,25 ± 0,06
P Value	0,364	0,502	0,100*	0,44	0,430
F Value	1,05	0,70	1,39	0,84	0,84

¹Means (n = 10) followed by the same letter do not differ by the Fisher test (P < 0.01–0.05–0.10)²Analysis of variance represented by capital letters compare the different treatments in the column³Control - animals not treated with Polypropylene; Treatment with 100 or 500 µg of Polypropylene

*significant for P < 0.10; **significant for P < 0.05; ***Significant for P < 0.01

exposed in aquariums with dosages of 10 and 100 mg/L of microplastic. Banaee et al. (2019) reported an increase in ALT in *C. carpio* after contact with polyethin microplastic at a dosage of 250 and 500 µg/L. The studies by Wen et al. (2018) demonstrated a significant increase in ALP in the *Symphysodon aequifasciatus* when exposed to polyethylene microplastic at a dosage of 25 mg/L.

A significant increase ($p < 0.05$) in serum cholesterol levels and a decrease ($p < 0.10$) in creatinine levels were observed in fish fed with 500 µg of polypropylene when compared to control animals (Table 2). According to Kim et al. (2021), the increase in AST could cause hormonal disorders, metabolic disorders of lipids, and lipoproteins caused by liver problems, a fact that would result in increased cholesterol and triglycerides in fish. Brandts et al. (2021)

observed a significant increase in cholesterol in the species *Sparus aurata* when exposed to PMMA microplastic (poly-methylmethacrylate) at a dosage of 10 mg/L, thus leading the animal to severe nutritional problems. Karami et al. (2016) points out the importance of the liver as the organ responsible for metabolizing lipids and reported how micro-plastics can affect its functionality, with a significant decline in this lipid in *C. gariepinus* when exposed to polyethylene microplastics at a dose of 500 µg/L. In this study, the values of triglycerides, creatinine, total proteins, and albumin did not show significant differences between treatments with polypropylene at 100 or 500 µg for *O. niloticus* (Table 2), corroborating the findings of Espinosa et al. (2017) who did not observe changes in total protein levels in *S. aurata*, which indicated normal osmotic balance in the fish after 30

Table 3 Mean values¹ ($\pm$ SE) and ANOVA² observed in the somatic index of spleen, heart, liver, stomach, caudal kidney, cranial kidney, gills and brain of tilapia fed with polypropylene

Treatment ³	Spleen (%)	Heart (%)	Liver (%)	Stomach (%)
Control	0,025 $\pm$ 0,0022	0,030 $\pm$ 0,002 ^B	0,350 $\pm$ 0,034 ^B	0,131 $\pm$ 0,008 ^B
100 μ g	0,028 $\pm$ 0,0028	0,043 $\pm$ 0,003 ^{AB}	0,471 $\pm$ 0,028 ^{AB}	0,179 $\pm$ 0,013 ^A
500 μ g	0,030 $\pm$ 0,0023	0,060 $\pm$ 0,006 ^A	0,520 $\pm$ 0,059 ^A	0,161 $\pm$ 0,010 ^{AB}
P Value	0,562	0,050**	0,093*	0,077*
F Value	0,58	3,09	2,48	2,68
Tratamento ³	Caudal Kidney (%)	Cranial Kidney (%)	Gills (%)	Brain (%)
Control	0,062 $\pm$ 0,005	0,077 $\pm$ 0,008	0,086 $\pm$ 0,006	0,092 $\pm$ 0,005
100 μ g	0,074 $\pm$ 0,006	0,077 $\pm$ 0,007	0,105 $\pm$ 0,008	0,094 $\pm$ 0,005
500 μ g	0,059 $\pm$ 0,006	0,071 $\pm$ 0,007	0,095 $\pm$ 0,006	0,100 $\pm$ 0,004
P Value	0,395	0,870	0,0327	0,700
F Value	0,94	0,14	1,14	0,36

¹Means (n = 10) followed by the same letter do not differ by the Fisher test (P < 0.01–0.05–0.10)²Analysis of variance represented by capital letters compare the different treatments in the column³Control - animals not treated with Polypropylene; Treatment with 100 or 500 μ g of Polypropylene

*significant for P < 0.10; **significant for P < 0.05; ***Significant for P < 0.01

Table 4 Mean¹ ($\pm$ SE) and ANOVA² values observed in the microbiological study of mesophiles and Gram-negative in the intestine of tilapia fed with polypropylene

Treatment ³	Mesophiles	Gram-negative
Control	11,92 $\pm$ 0,02 ^A	8,27 $\pm$ 0,47 ^C
100 μ g	10,62 $\pm$ 0,04 ^B	9,84 $\pm$ 0,14 ^B
500 μ g	10,23 $\pm$ 0,02 ^C	10,86 $\pm$ 0,02 ^A
P Value	0,001***	0,001***
F Value	794,69	21,31

¹Means (n = 10) followed by the same letter do not differ by the Fisher test (P < 0.01–0.05–0.10)²Analysis of variance represented by capital letters compare the different treatments in the column³Control - animals not treated with Polypropylene; Treatment with 100 or 500 μ g of Polypropylene

*significant for P < 0.10; **significant for P < 0.05; ***Significant for P < 0.01

days of exposure to 500 mg of PVC microplastic (polyvinyl chloride)/kg of b.w.

The hepatic, cardiac, and gastric somatic index (Table 3) revealed statistical differences with the increase of these organs in fish exposed to polypropylene. These alterations as a whole suggest the hypothesis of congestive circulatory alterations that would result in cardiac hypertrophy and impairment of the return circulation due to disturbances in the portal system. These findings corroborate the changes observed in serum AST and ALT activity, and the increase in the number of circulating leukocytes, possibly resulting in inflammatory changes in the stomach due to the presence of microplastics, which could alter circulation in the portal system and, consequently, cardiac function.

Another hypothesis raised by Browne et al. (2008), suggests that the presence of microplastic induces thrombosis in animals, causing damage to vascular tissues and cardiac

functionality of fish, corroborating the findings of the increased number of thrombocytes observed in tilapia. The average values observed in the somatic index for the caudal kidney, cranial kidney, gills, and brain did not demonstrate significant alterations between the animals submitted to the different treatments (Table 3).

The study of tilapia intestinal microbiota demonstrated a decrease in mesophilic aerobic microorganisms and an increase in Gram-negative microorganisms in fish exposed to the polymer, these findings being more significant in animals fed with 500 μ g of polypropylene (Table 4).

Changes in gastrointestinal tract immunity can result in significant variations in the intestinal microbiota, favoring certain groups of microorganisms over others (Belo and Charlie-Silva 2021). Tilapia treated with 500 μ g of polypropylene were positive for the presence of *Salmonella* spp., *Proteus vulgaris*, *Bacteroides* spp. and *Aeromonas* spp. which are Gram-negative microorganisms, in addition to being negative for the growth of Gram-positive microorganisms such as *Staphylococcus* spp., *Micrococcus* spp. and *Lactobacillus* spp. (Table 5). These alterations in the microbiota compromise the gastrointestinal functionality, resulting in disturbs in the digestive process, as well as in gut immunity, conditions that harm fish health.

In recent years, a significant number of studies have demonstrated the deleterious effect of MPs on aquatic organisms at different trophic levels, through experimental bioassays or surveys in the natural environment. Although these studies involve plastic polymers with different compositions, concentrations and forms of exposure (in water or food), it is observed in most of them that higher doses for prolonged periods result in more serious harmful effects, corroborating the findings of our study. It is worth mentioning that the

Table 5 Microorganisms observed in the count of Gram-negative and positive bacteria present in the intestine of tilapia treated with polypropylene

Microorganisms	Treatment		
	100 µg	500 µg	Control
<i>Salmonella spp</i>	Positive	Positive	Negative
<i>Pseudomonas spp</i>	Positive	Positive	Positive
<i>Proteus vulgaris</i>	Negative	Positive	Negative
<i>Bacteroides spp</i>	Positive	Positive	Negative
<i>Aeromonas spp</i>	Negative	Positive	Negative
<i>Staphylococcus spp</i>	Negative	Negative	Positive
<i>Micrococcus spp</i>	Positive	Negative	Positive
<i>Bacillus spp</i>	Positive	Positive	Positive
<i>Lactobacillus spp</i>	Positive	Negative	Positive

situation is alarming, in view of the continuous increase of plastic waste in the environment.

Conclusion

Tilapias fed daily with diets containing 100 or 500 µg polypropylene for 30 consecutive days showed deleterious effects, resulting in systemic inflammatory disturbs by altering liver functions, leukocyte profile, and organ morphometry, as well as changes in the intestinal microbiota. A dose-response effect was observed for these findings, which were more relevant in animals fed with 500 µg of polypropylene.

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Declarations

Competing Interests The authors declare that there is no conflict of interest regarding the publication of this article. The experiment design was approved by Ethics Committee on the Use of Animals of Brazil University under protocol n° 2200039.

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